

Effects of feeding different levels of dried cassava leaves (*Manihot esculenta*, Crantz) based concentrates with *Panicum maximum* basal on the performance of growing West African Dwarf goats

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panelA.O. Oni ^a, O.M. Arigbede ^b, O.O. Oni ^d, C.F.I. Onwuka ^a, [U.Y. Anele](#) ^b, B.O. Oduguwa ^c, K. O. Yusuf ^a

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Abstract

The feeding value of dried cassava leaves as supplement in the diet was investigated in an experiment with West African Dwarf goats at different inclusion levels of 0, 20, 40 and 60% of DM respectively, using [guinea grass](#) as a basal feed.

The study lasted for 112 days during which feed intake, performance characteristics, [digestibility](#) and nitrogen utilization were monitored in 40 male goats using a completely randomized design. Dry matter (DM) intake from guinea grass was significantly ($P < 0.05$; 0.01) increased from 0% to 60% inclusion of dried cassava leaves in the diet. The total DM intakes were 482, 526, 537 and 528 g/d for the 0, 20, 40 and 60% of dried cassava leaves in the diet, respectively. The body [weight gain](#) of the animals were significantly ($P < 0.05$) higher than in the control group. The highest gain was recorded in goats with 60% of dried cassava leaves in the diet (52.9 g/d) and lowest for the control (33.8 g/d). The inclusion of dried cassava leaves significantly ($P < 0.05$) decreased the apparent digestibility of DM, [neutral detergent fibre](#) (NDF) and acid detergent fibre (ADF), while it increased the apparent digestibility of crude protein (CP) and ether extract (EE), resulting in higher [nitrogen balance](#) (3.75 g/d) at 60% of the supplement. In conclusion, supplementing a basal diet of guinea grass (*Panicum maximum*) with dried cassava leaves improved DM intake, weight gain, apparent digestibility and N-retention in WAD goats. The inclusion level of dried cassava in the diet can be up to 60% level in a compounded ration for ruminant animals on DM basis.

Introduction

The productivity of small ruminants, especially goats, in many tropical countries is often poor, and has been related to disease, nutrition, genotype and management (Peacock,

1996). The limitations of ruminant nutrition in tropical and sub-tropical regions can at least partly be explained by the seasonal fluctuations in feed supply and pasture quality (Poppi and McLennan, 1995). From 1970, findings from the research of findings from several researches suggested that ruminant feeding systems based on poor quality tropical foliages, crop residues or agro-industrial by-products, in which protein is one of the first limiting factors, may require additional protein and roughage to maintain an efficient rumen ecosystem that will stimulate nutrient intake and improve animal performance. The increasing pressure on the use of cereals by the human population and livestock feed millers, coupled with a deficit of animal protein intake especially in developing countries necessitate the use of unconventional feedstuffs for livestock production. One such feedstuff is cassava which has been successfully incorporated into diets not only for ruminants but also for non-ruminants to replace maize completely. The most important use of cassava is as human food, serving as a primary, secondary or supplementary staple for over 200 million people in Africa. Cassava was also used as livestock feed, and regularly fed to sheep and goats on small-scale subsistence farms in Africa. Recently, interest has been focussed on foliage from cassava (*Manihot esculenta*, Crantz) as an animal feed. Cassava is an important food crop in tropical countries such as Brazil, Nigeria, Indonesia and Thailand. It is an annual root crop grown widely in tropical and sub-tropical areas. It can thrive in sandy-loam soil with low organic matter, receiving low rainfall and high temperatures. It is therefore, a cash crop cultivated by small-holder farmers within the existing farming systems in many countries (Wanapat, 1999). Cassava roots contain high levels of energy and minimal levels of crude protein, and have been used as readily fermentable energy in ruminant rations. Against the total world root production of 175 million tons, Nigeria produced 33,854,000 million tons (FAOSTAT, 2001). Large quantities of foliage are, however, available at the root harvesting and it is also possible to grow cassava as a forage crop (Preston, 2001). Cassava leaves, a by-product of cassava root harvest is (depending on the varieties) rich in protein (14–40% DM), minerals, vitamins B1, B2, C and carotenes (Adewusi and Bradbury, 1993). Cassava leaves have been used as a protein source when collected at the time of root harvesting. According to Wanapat (2003), harvesting of cassava at an early growth stage (3 months) to make hay could reduce the condensed tannin content and increase the protein content resulting in a higher nutritive value. Available literatures however, suggest that apart from lower methionine, lysine and perhaps isoleucine content, the amino acid profile of cassava leaf protein compares favourably with those of milk, cheese, soybean, fish and egg (Ayodeji, 2005). Despite these qualities, the nutritional potentials of cassava leaf meal and concentrates based diets remain currently under-researched in Nigeria. Recently the Federal Government of Nigeria launched a massive and aggressive campaign for the large scale production of cassava for export purposes, for the production of glucose syrup and

inclusion in wheat flour for bread baking. This has led to the production and distribution of the seedlings to farmers by government agencies and research institutes. Arising from this, it is envisaged that in not too long a time, this planted cassava will be matured for harvesting and processing. While the tubers will be processed for use for the above listed purposes, the stems will be replanted and the foliage left to waste on the farm. The potential of cassava as a grain substitute in livestock feed is yet to be fully exploited, as only a small proportion of total world production is currently being used, mainly in compounded non-ruminant diets. Ruminants can be fed not only cassava tuberous roots, but also the stem, leaves, and peel, and the various by-products of tuber processing. Since cassava foliage is known from previous works to be rich in protein despite the presence of cyanide, the foliage can be dried and included in a compounded ration for small ruminants. The present study is therefore, designed to determine the effects of feeding different levels of dried cassava leaves as supplement to guinea grass on West African dwarf goats.

Location and climate of the study area

The study was conducted in the Small Ruminant Experimental Unit, College of Animal Science and Livestock Production, University of Agriculture, Abeokuta, Nigeria. The experimental site was located in the derived savannah vegetation zone of south-western Nigeria. The climate in this area is tropical, with a wet season from March to October and a dry season from November to February. Annual rainfall averages about 1100 mm and the peak rainfall occurs in the period June–September. The temperatures

Results

The chemical composition of the experimental concentrate diets is given in Table 2. The DM was similar for all the diets while *P. maximum* had a dry matter content of 367 g/kg DM. The control diet (0%) and *P. maximum* had OM higher than 900 g/kg DM. The CP contents of the concentrate diets were significantly similar and increased from the control diet as the level of inclusion of cassava leaves in the diets increased from 173 to 176 g/kg DM. The crude protein content of *P. maximum* was less than half

Discussion

The high CP content (172.6–175.5 g/kg DM) obtained in the study involving feeding graded levels of dried cassava leaves to WAD goats was consistent with Ngo et al. (2005) but lower than the values of 200–260 g/kg reported by Mui, 1994, Wanapat et al., 1997, Van & Ledin, 2001. The content of CP depends on the proportion of the leaves and stem. In the present study, the petioles were included and this may have been the reason for the lower CP. The fibre fractions showed that the diets have the

Conclusion

The inclusion of dried cassava leaves at 60% promotes good growth performance, nutrient digestibility, nitrogen utilization in West African dwarf goats and lower feed costs. It is therefore, imperative that at smallholder livestock production system, the use of cassava leaves should be encouraged as animal feed instead of the present practice, which encouraged its use as green manure, mulch or even burnt in preparation for another planting season.